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Cost Optimization of Electrification for LNG and Oil and Gas Production Sites

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Abstract

Electrification is one of the main levers for the massive reduction in GHG emissions in LNG and oil and gas production sites. Many electrification projects have already been implemented and are currently under development. To enable more electrification projects, it is important to optimize their costs.

The overall cost and net technical cost (in \$/tCO₂) of dozens of electrification projects for onshore and offshore oil and gas production sites and LNG assets were analyzed. The projects were categorized based on their scope and net technical cost. A detailed cost analysis for each electrification scope was performed and the major cost contributors were identified. Based on this analysis a merit ranking of electrification scopes was established.

Ten levers to optimize the cost were identified: 1. Prioritize power reserve optimization 2. Electrification readiness 3. Competition 4. Qualification of more adequate equipment 5. Design specifications and redundancy 6. Standardization 7. Investment in R&D 8. Insurance and warranty 9. Project organization and 10. Minimize electrical infrastructure scope. Each lever has been detailed for major electrification projects.

This paper presents novel insights on cost analysis and cost reduction related to electrification projects for LNG and oil and gas sites. These insights are essential to enable the implementation of electrification projects and thus massively reduce GHG emissions.

Introduction

GHG emission reduction roadmaps often distinguish between scope 1, 2, and 3 emissions. The carbon intensity of oil, gas, and LNG production depends on various factors and is related to scope 1 and 2 emissions, which on average represent approximately 10% of the GHG emissions, while scope 3 typically accounts for about 90% [1]. This paper focuses on scope 1 and 2 emissions, with primary emphasis on scope 1. The main sources of scope 1 emissions in oil and gas production facilities include flaring, the combustion of fuel gas or liquid fuel, and venting (such as the release of native CO₂ from an acid gas removal unit). Fuel gas combustion is primarily used to generate electricity (via gas turbines or internal combustion engines), operate turbocompressors and turbopumps, and produce heat or vapor. The primary electricity consumers are typically compressors and pumps.

It is very challenging to significantly reduce emissions related to the use of fuel gas, particularly for brownfields, as they are an integral part of normal operations. There are many ways to achieve a moderate to considerable reduction of the gas fuel combustion emissions, such as the use of a Combined Cycle Gas Turbine (CCGT) and optimization of the gas turbines' power reserve [2]. For a substantial reduction in gas fuel combustion emissions three main approaches can be considered:

- i. **Electrification with green electricity:** this approach is the most mature but challenging due to the high capital expenditure (CAPEX) and operational expenditure (OPEX) associated with it. However, the unburnt fuel gas can often be valorized, improving the attractiveness of electrification. Optimization of the CAPEX is the topic of this paper.
- ii. **Carbon capture & sequestration:** this approach is challenging due to the low CO₂ partial pressure in the flue gas and the high CAPEX and OPEX associated with it. There is also no unburnt fuel gas to be valorized. On the contrary, this option requires even more energy.
- iii. **Use of low carbon fuels (green H₂, biofuels, etc.):** this approach is challenging due to the often very high quantities of low carbon fuels required to replace fuel gas, resulting in logistics challenges and very high OPEX. The unburnt fuel gas can often be valorized, improving the attractiveness of the use of low carbon fuels.

There are several articles and reports on the technical aspects of electrification [3], [4], as well as reports on its impact on greenhouse gas emissions and the intensity of oil and gas production [5]. For example, in 2022 the IOGP (The International Association of Oil & Gas Producers) published a technical report with recommended practices for electrification of oil & gas facilities [3]. Several major electrification projects have already been implemented, and, for example, the power from shore projects in the North Sea have significantly reduced the carbon intensity of oil production. In Norway values as low as 1.2 kg CO₂ per boe have been reported, for a fully electrified offshore asset connected to the national grid via a power from shore [5]. However, electrification projects are very CAPEX intensive, and it is essential to optimize their cost to further enable their implementation. While previous papers mainly focused on technical aspects, this paper aims to identify several levers to optimize the cost of electrification projects.

Technical aspects of electrification

Electrification is thus a major lever to reduce the GHG emissions of operations. One can distinguish four types of electrification projects:

- i. Connect only: the power generation unit is (partially) replaced by a connection to a national grid.
- ii. Convert only: the electrification of equipment (e.g., pumps, compressors, heaters)
- iii. Connect + convert: a combination of 1 and 2.
- iv. Electrical interconnections between sites, to mutualize power generation.

Figure 1 shows the main electrification bricks:

Technological bricks electrification

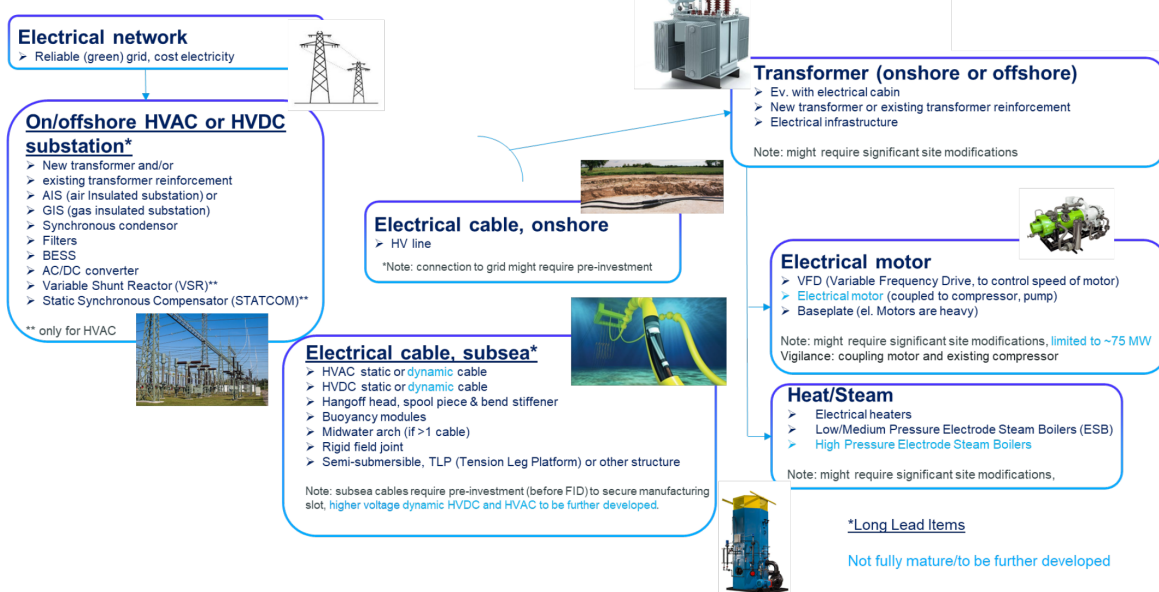


Figure 1—Technological bricks of electrification. Long lead items are underlined. Items that need further development are shown in light blue.

A **connect scope** primarily requires a reliable (national) electricity grid, with low-carbon electricity sources such as wind, solar, nuclear or hydropower. Grid stability is essential. One further needs a (new or modified) substation, onshore or subsea transmission, and transformers to transfer the electricity to the voltage required by the site. Both the substation and the subsea electrical cables are long lead items. Moreover, dynamic HVAC cables are still being qualified for higher voltages and dynamic HVDC cables still need to be developed.

It should be noted that heat is often generated by a Waste Heat Recovery Unit (WHRU) on the exhaust of the power generators. When connecting to the grid, an alternative way to generate heat should thus be provided. The following options can be considered:

- Partial connection to grid: continue to use the GTGs with WHRU for power and heat generation. Only replace GTGs without WHRU by grid connection. This option might be attractive in terms of CAPEX, but it should be noted that if the GTGs are operated at low load, just enough to generate the required heat with the WHRU, their efficiency might be very low, and the carbon intensity of the electricity will be high. The overall complexity of a connection to the grid should not be underestimated. It might for example include complex cyber security measures.
- Generate heat with an electrical heater or put an electrode steam boiler (ESB). This option has been retained for SnØhvit (LNG) connection to grid [6]. Note that the cost of the electrical infrastructure required to operate electrical heaters or ESBs might be several times the cost of the equipment itself. It is also important to have a clear view of the maintenance cost (replacement of electrodes). There have been failures with medium pressure ESBs in the past (with release of steam). High pressure ESBs still require further maturation as there is currently only one in operation [7].
- Generate heat with a boiler using natural gas.
- Install a new WHRU on another hot exhaust stream, for example on a compressor gas turbine. This option is often very expensive.

For a **conversion scope** (equipment electrical conversion), the following technical aspects are important:

- i. Compressors: Replacement of a gas or steam turbine by an electric motor.
 - o Gas and steam turbine: the replacement of only the turbine by an electrical motor has been done in refining (steam turbine) but has not been done on LNG plants (gas or steam turbine) at the time of writing. The coupling between the motor and the compressor needs to be precisely calculated to avoid strong vibrations. There is a high risk associated with this. Stop of the compressor to replace incorrect coupling has already been observed. Moreover, the shutdown time might be long (structure and foundation not adequate for weight motors, and so on) [8], [9].
 - o Gas turbine: For relatively small compressors it is often more attractive to replace the gas turbine and the compressor with hermetic compressors (e.g., Integrated Compressor Line, ICL) instead of only replacing the gas turbine with an electrical motor. The reason is the shorter shutdown, the lower risk related to coupling motor- compressor and similar cost.
- ii. Electrode Steam Boiler: See comments above.

Analysis of electrification project costs

Dozens of electrification studies have been analyzed. The results are shown in Figure 2. The Net Technical Cost (NTC, in \$/tCO₂ avoided) is given as a function of the total CAPEX (all converted into million US \$2024) for different electrification scopes (not necessarily the entire electrification project). The NTC is the CAPEX+OPEX-fuel gas valorisation divided by CO₂ avoided. The colours correspond to different electrification scopes. The values are estimates with varying accuracy due to the combination of feasibility studies, conceptual studies, FEED (Front End Engineering & Design), and completed projects. The size of the circle represents the GHG emissions savings over the project duration, which is typically 20 years, but not always (for example if there is an end of production before 20 years). Figure 2 is useful for a first merit ranking of electrification scopes.

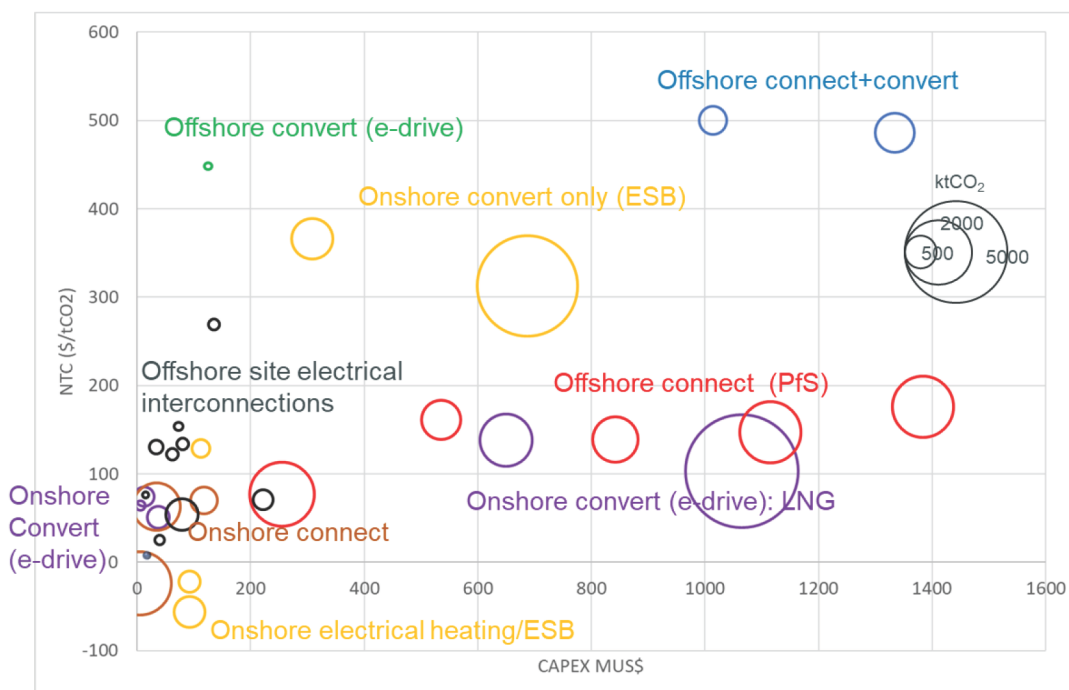


Figure 2—The Net Technical Cost (NTC) (in \$/tCO₂ avoided) and the total CAPEX (in million US\$2024) for dozens of electrification projects. The size of the circle is proportional to the total amount of avoided CO₂ emissions for the entire project duration.

There are three types of electrification scopes with an **NTC above 200 \$/tCO₂**. These projects are the most expensive projects.

- i. **Offshore connect plus convert:** projects connecting existing offshore platforms to a national grid (by power from shore) and replacing the main machines' gas turbines (compressors, pumps) to e-drive (convert). This is especially not interesting for small, single platforms.
- ii. **Offshore convert only:** for example, replacing a gas turbine of a compressor by an electrical motor on an offshore platform, with power coming from existing GTGs.
- iii. **Onshore convert only (ESB):** installing large electrode steam boilers, requiring significant electrical infrastructure.

There are three types of electrification scopes with a **NTC between 100 and 200 \$/tCO₂**. Those projects should be a priority for cost optimization:

- i. **Offshore connect only** (power from shore, PfS): connecting an offshore platform to a national power grid via a power cable from shore, without any conversion of machines on the offshore platform.
- ii. **Onshore convert (e-drive): LNG:** replacing a gas turbine of a compressor with an electrical motor in an existing LNG plant.
- iii. **Offshore site electrical interconnections** (in some cases).

And finally, there are electrification scopes with a **NTC lower than 100 \$/tCO₂**. Those projects are the most attractive ones in the merit ranking.

- i. **Onshore connect:** connection of an onshore oil & gas or LNG site to the national grid, without any conversion.
- ii. **Onshore convert (e-drive):** replacing a gas turbine of a compressor with an electrical motor in an existing oil & gas plant. In practice often the entire compressor is replaced with an ICL.
- iii. **Onshore electrical heating/ESB:** if the electricity is very cheap and/or if the project is relatively small with a limited electrical infrastructure scope.
- iv. **Offshore site electrical interconnections** (in some cases).

In a second step, the detailed cost breakdown of the different electrification architectures was analyzed. The main conclusion is that the principal cost element is case-dependent. For example, when replacing a compressor gas turbine with an electric motor on an onshore oil and gas platform, the main cost element is the procurement of the electric motor (or ICL). For the same scope on an LNG asset, the main cost element is the installation and construction cost. For power-from-shore projects involving electrical interconnections, the main cost element is the procurement of the cable, the price of which has significantly increased in recent years. For the installation of large ESBs the main cost element is the electrical infrastructure.

Ten levers to reduce the cost of electrification projects

The main obstacle to implementing electrification projects is their high cost. Therefore, ten levers to optimize the cost of electrification projects were identified.

Prioritize power reserve optimization over electrical interconnections

For the specific case of a group of several related oil and gas production sites (e.g., several FPSOs, several platforms, and so on) where each site has its own power generation, it is often envisaged to mutualize the power generation by interconnecting the sites. Site electrical interconnection is CAPEX intensive. It is therefore strongly recommended to first optimize the power reserve of each individual site [2]. A methodology for power reserve optimization has been developed and deployed by TotalEnergies [2]. Power

reserve optimization often doesn't involve any CAPEX. When the power reserve of each site is fully optimized, one will in most cases conclude that the electrical interconnection between the sites is no longer beneficial. Power reserve optimization thus allows to avoid CAPEX intensive electrical interconnections.

For greenfield: electrification-readiness

For greenfield of all types that are not connected to the grid it is recommended to be all-electric. This means that there is a central power generation unit (PGU, with a high efficiency) and that all equipment is using this electric power. For offshore oil and gas platforms this is the main point, however, for LNG plants electrification readiness is less straightforward, as explained below:

Greenfield LNG: electrification-readiness.

For future connections to an electrical grid

When the power generation unit (PGU) is replaced by a connection to an electricity grid, one needs to manage the fuel gas balance differently. There could be excess fuel gas that was not anticipated in the original design. This should thus be anticipated by considering this case as a separate case during engineering studies (mass balances, equipment list, (additional) plot space, etc.)

For some gas reservoirs, the N₂ content in the produced gas increases with time. If this is the case, and if the GTG are replaced by a connection to grid, the N₂, which is normally evacuated via burning fuel gas in the GTG, might accumulate in the fuel gas. There are several solutions for which plot space and tie-ins should be foreseen [10]:

- i. End flash gas
- ii. N₂ stripping column
- iii. Nitrogen Rejection Unit (if grid connection and e-drive, e.g. Marsa LNG) (Note: complex retrofit)

Those require extra power for the compression of the fuel gas (booster compressor typically 5-10 MW) to be foreseen in PGU design or electrical infrastructure if power is imported.

For future replacement of compressor gas turbines by electrical motors:

Important general information:

- 75 MW electrical motor is maximum field proven today. 90 MW motors (same level of power than a Frame 7 gas turbine) are not in operation today.
- Transformer+VSD+electrical motor cost 5-8% more than a gas turbine, and one needs to add electrical infrastructure cost.
- Electrical motors are ~2 times heavier than gas turbines.

If a greenfield LNG plant has compressors driven by gas turbines, but would like to be ready for a future switch to electrical motors to drive the compressors following needs to be addressed:

- i. Depending on how much additional MW will be required, one will need to expand the electrical network, most likely at a higher voltage. This has a large footprint. One thus needs to foresee additional space or reservations:
 - Space for electrical equipment: an additional substation and infrastructure (AIS or GIS, harmonic filters, synchronous condensers, electrical room, air coolers for VSDs, transformers, etc.)
 - This space should not be too far from the electrical motors/VSDs.
 - The transformer and VSDs should not be too far from the motors (<300m).
 - Ideally all equipment is outside the compressor battery limit, to limit shut-down time.

- o Foresee reservations for future electrical cables: space in the cable trays or trenches for electrical cables (under or above ground). Cables are very heavy, so one needs to anticipate if a pipe rack is used or underground cables (civil works).
- ii. Electrical motors are ~2 times heavier than GT. One should thus foresee adequate structures/foundations (table top or grounded (plate) configuration for compressor).

For other recommendations such as the coupling between motor and compressor, see section project organization, replacement of compressor turbines by an electrical motor.

For future electrification of heat or steam generation

Heat is often provided by WHRU on GT exhaust of PGU and/or compressors, either directly such as for regeneration gas, or indirectly via a hot oil system. Electrification might thus impact heat generation.

If little heat is required (e.g., US LNG Plants: low native CO₂ content in the gas, no LPG) one needs to foresee plot space and electrical infrastructure for a heat pump and an electrical heater. There is a preference to keep the hot oil loop (don't switch to steam).

If much heat/steam is required (e.g., LNG plants with 2-3% native CO₂ in the gas, LPG (need steam for reboilers), regeneration molecular sieves) one needs to foresee plot space for electrical infrastructure and electrode steam boilers or electrical heaters. Heat Pumps can also be proposed. One also needs to foresee water management (conductivity control) because ESBs require high quality water.

Competition is key to reducing costs

To reduce the cost of electrification projects it is important to introduce (qualify) new suppliers. Qualifying new suppliers will not only create a competitive environment, driving down the prices, but might also improve the quality of products and services, promote innovation and diversification, ameliorate supply chain resilience, and so on. Another advantage is getting production slots on our terms. This is particularly true for electrical subsea cables.

An alternative approach would be to consider a strategic partnership with a supplier, but this is only recommended if there is a strong technological advantage.

Qualification of new suppliers is particularly important for subsea electrical cables, and especially for dynamic cables as well as for large electrical motors and electrical heaters/boilers.

Qualify more adequate equipment

This is closely related to the introduction of new suppliers.

A typical example is the subsea cable for a power from shore project. Depending on the electrical demand and the distance, one might need to lay several qualified lower voltage cables in parallel because the single higher voltage cable is not yet qualified. Qualifying the higher voltage cable would significantly reduce the electrification cost, as laying one cable is cheaper than two or more cables in parallel.

For electrical equipment like transformers, it is recommended to qualify more compact design options. The weight reduction directly translates into a cost reduction, especially offshore.

Define appropriate design specifications, reduce overdesign, challenge redundancy

Today, many electrification project scopes are relatively new or have not been frequently implemented. As a result, design specifications from similar project scopes are often used. It is important to develop design specifications for specific electrification scopes to ensure that the selected components are appropriately sized and rated for their intended use. Overdesign occurs when components are specified with higher capacities or features than necessary, leading to unnecessary costs and inefficiencies.

Examples of the reduction of overdesign include accepted cable heating and insulation, and wet design instead of lead sheath for static cables. This kind of optimization can reduce costs by approximately 5%.

Equally important and related is the questioning of redundancy. Redundancy should systematically be challenged, for new equipment, but also for new versus installed equipment. An example of the latter is a connection to grid to replace the GTGs: one should question whether it is really required to keep one or more GTGs in operation or in standby when a connection to the grid is established.

For cables, one should avoid routes in locations with more stringent specifications (fishing areas, protected coastlines).

Standardize

Standardizing equipment means using the same design, specifications, and components across multiple applications. This reduces the need for custom designs and ensures consistency in the equipment used. Standardization simplifies the procurement process, as the same equipment can be ordered in bulk, often at a lower cost due to economies of scale. It also simplifies maintenance, and it allows us to mutualize spare parts and thus reduces the need for a large inventory of different spare parts. It might also improve reliability.

Typical examples include transformers, small electrical-driven compressors, and so on.

Invest in R&D

Even though electrification is a major lever to reduce GHG emissions, R&D efforts are often limited compared to other major decarbonization levers like CCS or decarbonized fuels. This is primarily because electrification is technically much more mature. However, its cost is significant, and more R&D efforts are required to lower the costs of electrification projects. An example is subsea cables. They are currently extremely expensive, and more research and development is required to identify new cable types, (insulation) materials, subsea cable interconnection systems, disruptive cable technologies, monitoring systems, and so on, that can optimize costs.

Insurance and Warranty

Insurance policy is a relatively complex topic that cannot be generalized. The purpose of this section is to provide some key elements.

There are basically two types of insurance: property damage coverage and civil responsibility insurance. The property damage coverage is an operational coverage. It is multi-risk, and covers assets (incl. subsea, inventories, and so on.) for various damages. Normal wear and tears are not insured. Insurance intervenes only when there is an external event (e.g. a fire). Insurance can be quite expensive depending on many factors, the country of operation among others. Energy companies typically focus on the risks of large losses. The costs related to recommissioning are typically included, but production losses are often not insured, as well as the defective part itself.

To reduce insurance costs one can either reduce the scope of the insurance or increase the insurance deductible. Construction activities related to maintenance are covered by a specific section.

For extraordinary operations there is also the CAR insurance (Contractors All Risk insurance), suitable for construction activities that are more risky than standard operations. Cable laying (incl. tracing phase, preparation) for example would be typically covered by CAR. The CAR can be taken by the energy company or, alternatively, one could be insured via the cable laying company, or via a combination of both (e.g. the energy company could ask the cable laying company to contribute a certain amount of the deductible if they are responsible for the damage). During the construction phase the defective part can be included. Once the electrical connection is established, the cable falls under the operational coverage.

The insurer is typically an external company, unless the company is self-insured.

The evolution of the insurance costs depends on the claim rate. In some sectors it is increasing very fast, for example in offshore wind, due to the many natural disasters (wind) and due to cable damage.

The risks should thus be assessed properly, on a case-by-case basis. It is important to clearly understand how the insurance costs are calculated when included in the overall procurement cost of the equipment.

The insurance cost is also related to the measures taken to reduce the risks. For example, one can work with the same cable supplier and cable layer. Or one can invest more in cable protection (e.g. cable mattress), especially in areas near the shore, but this will increase the CAPEX of the project. An optimum needs to be found to reduce the overall cost of cable protection and insurance by working with suppliers, bankers, and insurance companies to manage risks. This includes avoiding routing cables in locations with stringent specifications, such as fishing areas or protected coastlines.

An equally important aspect of electrical cable insurance is civil liability (damage to third party). It is important to properly assess and insure this risk.

A warranty is provided by the equipment manufacturer. The duration of the warranty has an impact on its cost and depends on the request of the customer (typically 24 months). The warranty is related to the technical specifications of the equipment. It often takes effect from the moment of installation, sometimes from the moment of delivery. For subsea cables, for example, it is challenging to include the laying of the cable in the warranty.

Project organization

It is important to find the right balance between dividing the project into separate work packages (the advantage of doing so is maintaining competition) and maintaining flexibility (too many work packages complicate the project organization).

The planning should be such that work is done as much as possible before the shutdown.

The following recommendations are specifically for the electrification of compressors (see also section Electrification readiness):

- i. It is recommended to envisage to work with modules. For example, for an offshore compressor electrification project it is recommended to optimize the size and weight of the CPM module (Compressor Power Module) containing all electrical equipment (VSD, transformers, cabinets).
- ii. Full replacement of compressor and drive (e.g. by ICL, technically limited to < 15MW) is often preferred for the following reasons:
 1. Replacing only the turbine by a motor will result in a higher destruct scope and in a longer compressor unavailability (major civil works, turbine dismantling, new motor integration, new lube oil system) and thus loss of production. A full replacement avoids this because one can install the new compressor while continuing to produce with the old compressor and just have a short stop to switch compressors.
 2. Replacing only the turbine by a motor requires more compressor maintenance compared to an ICL (dry gas seals source of unavailability, seal leakage risk).
 3. Replacing only the turbine by a motor requires a dedicated engineering study for the compressor-motor coupling. This coupling is a major risk and should be well calculated to avoid, for example, compressor limitations due to vibrations. To limit this coupling risk, it is recommended to acquire the motor from the same company as the compressor, but this limits competition and might thus increase the cost.
 4. Replacing only the turbine by a motor shows a higher risk of cost and schedule escalation.
 5. The new compressor might be better adapted to the present rates and thus have a lower energy consumption.
 6. Although the equipment cost of an entire new compressor is often higher than just a new motor, when contingencies are added as well as the considerations above, the new compressor option is often cheaper overall.

Note: Motors are often a bespoke design. In the event of failure of the motor, full replacement is very likely. Therefore, a critical decision is the level of sparing for the motor included in the initial PO (Purchase Order).

- iii. Before the PO of the main equipment, it is important to discuss in detail all the requirements to avoid expensive changes after the PO.
- iv. The cable scope should not be underestimated and should be considered at the earliest stage, similar to the piping scope:
 - o underground: there might be pollution, very complicated.
 - o above ground: need extra support on cable trays. Huge pipe racks due to the very heavy weight of cables.
 - o consider using more cables with a small diameter rather than one cable with a large diameter, as they are much easier to handle.
- v. Adequate planning is required. It is recommended to first determine the economically acceptable period, then plan accordingly. The inverse approach might result in too long a project duration and much higher costs.

Select equipment that minimizes equipment and electrical infrastructure cost

In electrification projects the electrical infrastructure cost can be much more important than the equipment cost. This is for example the case for large electrode steam boilers. One should therefore avoid selecting the cheapest equipment, without consideration for the electrical infrastructure. One should select the equipment that minimizes both equipment and electrical infrastructure costs.

Conclusions

Electrification is a way to significantly reduce greenhouse gas (GHG) emissions in LNG and oil and gas production sites. By optimizing the costs of electrification projects, it is possible to make these initiatives more attractive. The TotalEnergies electrification taskforce has identified ten key levers to achieve cost optimization: 1. Prioritize power reserve optimization 2. Electrification readiness 3. Competition 4. Qualification of more adequate equipment 5. Design specifications and redundancy 6. Standardization 7. Investment in R&D 8. Insurance and warranty 9. Project organization and 10. Minimize electrical infrastructure scope. Each lever has been detailed for major electrification projects. These strategies have been illustrated for various projects.

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